

GMP 1

Science

May, 2026



**SE
TU**

Ollscoil
Teicneolaíochta
an Oirdheiscirt

South East
Technological
University

Short Title:	GMP 1	
Faculty	Science and Computing	
Department	Science	
Cluster	Quality, Regulatory and Compliance	
Author	BWHELAN	
Credits	5	Level: Introductory

Description of Module / Aims

This module covers all aspects of current Good Manufacturing Practice (cGMP), the aim of which is to ensure that pharmaceutical products are safe, effective, pure and free of contamination. The student will also learn how to interpret guidelines for quality systems and they will be able to assist with inspections of the manufacturing environment.

Programmes

GMPT-0001 Higher Certificate in Science in Good Manufacturing Practice and Technology 1 / 1 / M
(WD_SGMPT_C)

Indicative Content

- Definition of a medicinal product; role of the Competent Authorities in Europe and USA; associated pharmaceutical legislation; regulatory inspections
- Quality assurance principles: establish quality characteristics; auditing procedures to manage Quality Systems
- Validation life-cycle; Design Qualification, Installation Qualification, Operational Qualification, Performance Qualification; Change Control to maintain the validated system
- Management of customer complaints, as a means to improve process control; performance metrics; CAPA reports
- Control of documentation; Standard Operating Procedures (SOPs); Batch records; software packages for electronic documentation

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify the function of Good Manufacturing Practice in the production of medicinal products.
2. Complete quality audits.
3. Construct a validation protocol with acceptance criteria to prepare a validation report.
4. Apply corrective and preventative actions in a manufacturing environment.
5. Record a sample Standard Operating Procedure and equivalent documentation.

Learning and Teaching Methods

- Lectures.
- Class discussion.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Assignment</i>	50%	1,2,4
<i>Case Studies</i>	50%	3,5

Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out practical/professional tasks.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical/professional tasks.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out practical/professional tasks.
- 60%-69%:** Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design when carrying out practical/professional tasks.
- 70%-100%:** Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on practical/professional tasks.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture		12
Independent Learning		123

Essential Material

- "European Medicines Agency." <http://www.ema.europa.eu/ema/>
- "Food and Drugs Administration." <http://www.fda.gov/>
- "Healthcare Products Regulatory Authority." <http://www.hpra.ie/>
- Ng, R. *Drugs: from discovery to approval*. 3rd ed. London: Wiley Blackwell, 2015.